Solution Architecture

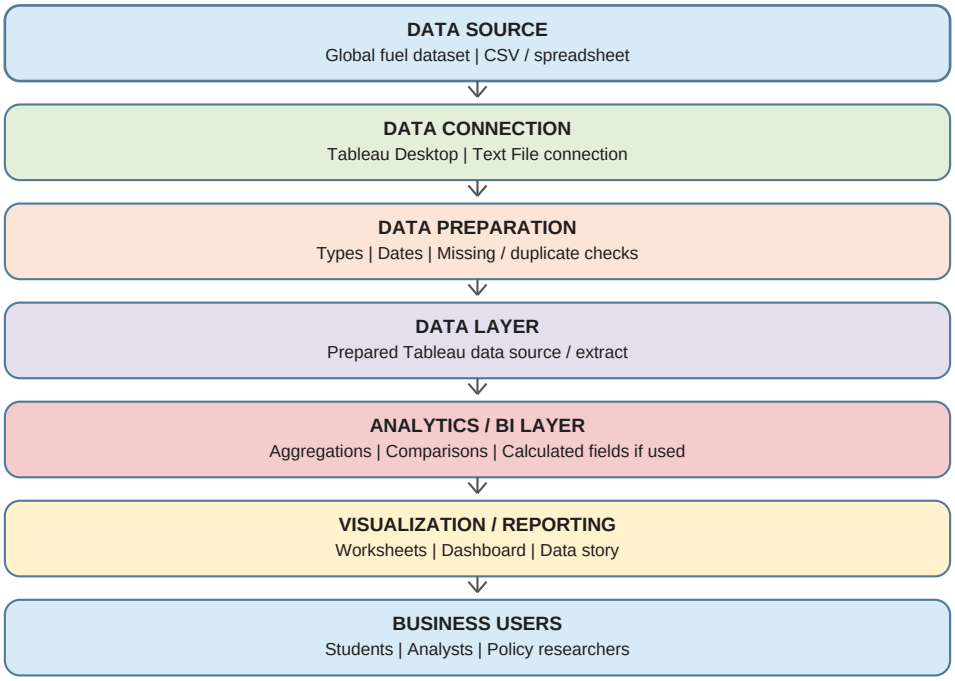
Project Detail	Information
Date	29 October 2026
Team ID / Enrollment No.	EN25CS3010450
Project Name	Global Fuel Prices and Taxation Trends
Maximum Marks	4 Marks
Project Type	Individual Project

Solution Architecture:

Solution architecture describes how the data source, preparation steps, analytical logic and visualization components work together to address the project's analytical problem. This Tableau workflow moves from the supplied fuel-price dataset through data preparation and analysis to visual reports for users.

Its goals are to select a suitable analytics workflow, describe component roles and technologies, show how prepared data becomes visual insights, and document the solution structure.

Solution Architecture Diagram



Component Description Table

Component Name	Description / Role in Architecture	Technology Used
Data Source	The global fuel prices dataset containing available date, country, region, petrol price, diesel price, LPG price, Brent crude price, tax percentage, income level and subsidy level fields.	CSV / spreadsheet dataset
Data Connection	Connects Tableau to the source dataset so records can be loaded for analysis.	Tableau Desktop data source connection (Text File / CSV)
Data Preparation	Checks field types, date formats, missing values, duplicate records, category labels and unusual price values before analysis.	Tableau Data Source tab and worksheet-level data checks
Data Layer	Provides the prepared dataset used by worksheets, dashboards and the Tableau story.	Tableau data source / extract, depending on the saved workbook
Analytics / BI Layer	Summarizes and compares fuel-price measures and tax percentages by available region, country, date, income level and subsidy level.	Tableau aggregations, filters and calculated fields where used
Visualization / Reporting	Presents comparisons and trends through Tableau worksheets, a dashboard and a data story.	Tableau Desktop worksheets, dashboard and story
Business Users	Students, energy-market analysts, policy researchers and other readers explore the views to understand patterns in the dataset.	Tableau workbook (.twb / .twbx) and exported report views

Implementation note: This architecture reflects a CSV-based Tableau analytics project. It does not assume a separate database server, automated refresh, online publishing, or specific filters/calculated fields unless these are actually configured in the final workbook.